



ROCKET: Body & Feed System Group 5C

Propellant feed system, structure, and full body integration



Project Objectives & Specification

- This super-project developed and tested a **reusable liquid bi-propellant rocket system** in collaboration with Imperial College London Rocketry (ICLR), **targeting 10 kNs of impulse** for 10 hot fire cycles.
- As the **Body & Feed team**, we delivered a **fully integrated structure** using a **self-pressurising piston tank** and **internal piping**, feeding ethanol and nitrous oxide through the TVC stage to the engine at required conditions.
- Our main **targets** were **mass reduction** and **encapsulation of the body and piping within a uniform diameter**, improving upon last year's DMT Rocket.

Piston, Bulkheads & Sealing

This **mass-optimised** design involves a **single tank** made from aluminium, partitioned by a hollow **moving piston** sealed with **Viton O-rings**.

- Nitrous oxide** at **20°C** exists as a two-phase liquid and gas at **50 bar**, meaning the tank is **self pressurised**.
- The steel piston acts as a **barrier** and **transmitter of force**, pushing the ethanol out the top.
- Aluminium bulkheads serve sealing and structural purposes, **replacing last year's tie-rod design**.

The **nosecone**, following **Von-Karman** design, is bolted onto the top bulkhead. It holds space for a **potential payload**.

Top Bulkhead

Ethanol (**fuel**) above the piston and nitrous oxide (**oxidiser**) below are **fed through two lines** connected with **flexible piping** to the engine inlet.

The **stringer**, shown sectioned, encapsulates the piping and serves as attachment for TVC.

The **dip tube** serves as a stopper for the piston: they are **not** attached. The drilled hole serves as a filling indicator, marking target liquid nitrous volume.

Fuel raceway pipe connecting to the **main ethanol line**

The **main nitrous line** is connected through a fitting in the bottom bulkhead. A T-junction separates the line into a servo valve and a check valve used for filling.

The **nitrous vent line** connects via the dip tube. It includes a pressure transmitter and servo valve for regular venting, as well as a safety relief valve and burst disk for over-pressure protection.

Servo Valve

Pressure transmitter (PT)

The **main ethanol line** includes a PT to measure pressure, and a servo valve to control flow. It connects via the raceway pipe running along the tank.

Main Body Hydrostatic Pressure Tests

- Verified **structural integrity** and **sealing** under worst-case loads, to **avoid rupture and pre-mixing** of propellants
- Tank Pressurised to **80 bar for 60 minutes**, exceeding:
 - Operating pressure by 1.6x
 - Activation thresholds of safety devices
- Tank maintained sufficient pressure over 60-minute period **without** sustaining any **visible damage or leaking**.
- Digital Image Correlation used to **validate stress predictions**
- Sealing** across bulkheads and piston determined as **sufficient**.

Hot Fire & Cold Flow Tests

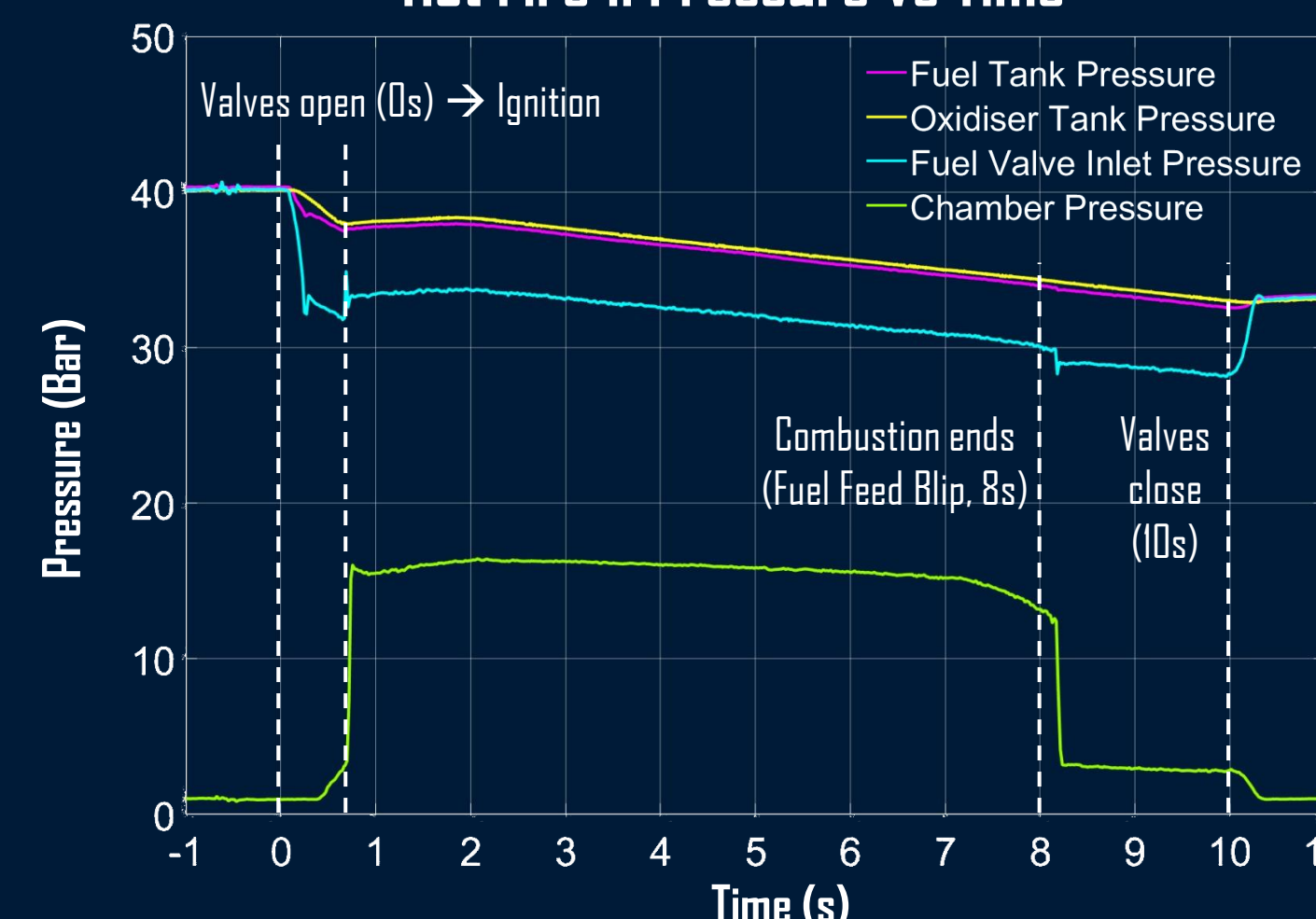
Cold flow: detected no leaks, verified valve operation & PT functionality, confirmed delivery of propellants to the engine.

- Nitrogen and distilled water simulated propellants

Hot Fire: pressure measurements verified piston's smooth operation

- 2 Hot Fires** performed, both with successful combustion
- 10 second** target achieved in second fire
- Fuel mass flow rate **78% of the target value**.

Hot Fire I: Pressure vs Time



Conclusion / Future Development

The integrated feed system, contained within a uniform diameter, was validated through two successful hot fire tests. In future DMT rocket iterations, development should focus on:

- Enhancing data acquisition during hot fire testing
- Improving ease of assembly accessibility in design.

These refinements pave the way for a scalable assembly as a student or demonstrator setup, with clear potential for future adaptation into a **flight-ready rocket**.